Xin Meng 孟昕

2027 PhD applications | 3D Medical Imaging · Multimodal Medical AI · Trustworthy Evaluation

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Research Profile

I am an MLIS student at the School of Health Management, China Medical University, conducting research in the Department of Clinical Epidemiology at Shengjing Hospital. I study **3D medical imaging, multimodal medical AI, and model reliability** under distribution shift through external validation, case-level failure analysis, test-time update audits, and evidence-bounded reasoning.

Education

China Medical University 2024.09 – 2027.06

Master of Library and Information Science (MLIS), in progress | Research focus: Medical AI | Shenyang, China

Jilin University 2018.08 – 2022.06

B.Mgmt. in Archival Science; B.Econ. in International Economics and Trade | Changchun, China

Research Focus and Methods

- **3D medical imaging:** Lung CT risk, longitudinal registration, patient splits, external validation, calibration, and matched CNN/Mamba/attention controls on LNDb and LIDC-IDRI; dual-teacher distillation analysis.
- **Multimodal medical AI and RAG:** Image–text fusion, text provenance, guideline retrieval, citation checks, abstention, and rule–language separation.
- **Trustworthy evaluation:** Cross-domain failures, adaptation benefit–risk, patient-clustered bootstrap, selective referral, and reproducible audits.

Accepted Papers and Presentations

† denotes equal first-author contribution.

Auditing Clinical Text Leakage and Text Dominance in Multimodal Lung Nodule Assessment 2026 · Accepted

Xin Meng†, Xiaopeng Li†, Huixin Yu†, Yuhong Zhao, Qing Chang | IEEE BIBM 2026 AIBMMD Workshop

My role: Defined minimal, safe, leaky, and de-leaked text conditions; compared text-only, image-only, and fusion models to separate leakage-driven performance inflation from text dependence; used patient-clustered bootstrap to estimate uncertainty.

Mining Multi-Model Failure Patterns for Reliability Auditing of 3D Lung Nodule Malignancy Predictors 2026 · Accepted

Xin Meng†, Xiaopeng Li†, Huixin Yu†, Yuhong Zhao, Qing Chang | IEEE ICDM 2026 ARRL Workshop

My role: Built the case-level failure-mining and selective-referral pipeline across seven 3D models; compared shared errors, high-confidence mistakes, and residual risks after referral.

A Benefit–Risk Audit of Test-Time Update Budgets for Longitudinal Medical Image Registration: A GrowthMamba Case Study 2026 · Oral presentation

Xin Meng†, Xiaopeng Li†, Huixin Yu†, Qing Chang, Yuhong Zhao | 32nd Chinese Medical Association Conference on Medical Informatics (paper in Chinese)

My role: Compared frozen inference with multiple test-time update budgets and jointly analyzed image similarity and Jacobian folding risk; under cross-domain stress, frozen inference achieved higher mean similarity than three online-update variants.

Trends and Hotspots in Deep Learning-Assisted Medical Imaging Research 2025 · Accepted

Xin Meng, Huixin Yu, Qing Chang, Yuhong Zhao | 31st Chinese Medical Association Conference on Medical Informatics (paper in Chinese)

My role: Retrieved relevant Web of Science Core Collection records from 2016–2025 and used VOSviewer to analyze keyword co-occurrence, citations, and author collaboration.

Selected Manuscripts Under Review

Auditable Decision-Invariant Retrieval-Augmented Lung-RADS Report Realization: A Verified-Subset Technical Evaluation 2026 · Under review

Xin Meng†, Xiaopeng Li†, Huixin Yu†, Yuhong Zhao, Qing Chang | IEEE JBHI

My role: Designed a pipeline where deterministic rules own Lung-RADS categorization and retrieval and the LLM only realizes fixed evidence and wording; audited citation validity, abstention, text leakage, and unsupported generation.

Auditing LLM-assisted scholarly reading: Replicate-session agreement in claim-span selection and evidence coding 2026 · External review

Xin Meng†, Xiaopeng Li†, Huixin Yu†, Yuhong Zhao, Qing Chang | Computers in Human Behavior Reports

My role: Designed the reproducibility audit for LLM-assisted reading and completed the methods, experimental software, statistical analysis, and visualization; compared claim-span selection and evidence-coding agreement across repeated sessions.

When "Not Documented" Is Not Clean: A Lineage-Aware Audit and Controlled Stress Test of CT Pretraining 2026 · Conference review

Xin Meng, Wenhao Zhu

My role: Audited direct and transitive relationships between CT pretraining corpora and evaluation sets for each configuration; tested representation contamination under controlled changes in pretraining scale and random seed, retaining clear limits on negative findings.

General-Purpose Vision-Language Models versus Task-Specific Models for Pulmonary Nodule Malignancy Prediction 2026 · Under review

Xin Meng†, Xiaopeng Li†, Huixin Yu†, Yuhong Zhao, Qing Chang

My role: Compared Gemini with seven task-specific 3D models on the same patient-disjoint LUNA25 holdout; evaluated CT-only, text-only, and combined inputs and used patient-clustered bootstrap to estimate uncertainty.

Additional co-authored manuscripts under review cover health policy text mining, Mendelian randomization, nutrition informatics, and public-health data analysis; the full list is at mengxinlab.github.io.

Research and Data Experience

National Health Commission Health Development Research Center 2025.11 – 2026.02

Remote research assistant, Prof. Wudong Guo's group; harmonized multi-province off-label drug-use consensus documents and group standards into a structured comparison table

Shanghai Zhizhongda Information Technology Co., Ltd. 2024.05 – 2024.08

Data analyst; used SQL/Tableau to analyze quotations, orders, and competitor prices in support of pricing rules

Gold Hongye Paper Group (APP / Qingfeng) 2023.04 – 2024.05

Data analysis management trainee; analyzed prices and promotions across five regions and 100+ core SKUs

Technical Skills and Academic Exchange

- **Programming and modeling:** Python, PyTorch, pandas, SQL, Git; 3D CNNs, Mamba, Transformers, knowledge distillation, multimodal fusion, and RAG.
- **Medical imaging and evaluation:** CT preprocessing, HU normalization, voxel spacing, patient-level splitting, external-cohort validation, calibration, subgroup and failure-mode analysis.
- **Academic exchange and credentials:** *Beyond the Leaderboard*, a medical-AI science-communication film, screened at the 32nd Chinese Medical Association Conference on Medical Informatics (2026); Science Communication Pioneer, 31st conference competition (2025); Information Systems Supervisor, National Intermediate Professional Certificate.
- **Languages:** IELTS 6.5 (Reading 7.5, 2024); CET-6 502.